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AS 2305: Substantive Analytical Procedures

Amendments: [Amending releases and related SEC approval orders](#)

Guidance on AS 2305: Staff Audit Practice Alerts [No. 8](#), [No. 12](#), and [No. 15](#)

.01 This section establishes requirements regarding the use of substantive analytical procedures in an audit.

Note: AS 2110, *Identifying and Assessing Risks of Material Misstatement*, establishes requirements regarding performing analytical procedures as a risk assessment procedure in identifying and assessing risks of material misstatement.

Note: AS 2810, *Evaluating Audit Results*, establishes requirements regarding performing analytical procedures as part of the overall review stage of the audit.

.02 Analytical procedures are an important part of the audit process and consist of evaluations of financial information made by a study of plausible relationships among both financial and nonfinancial data. Analytical procedures range from simple comparisons to the use of complex models involving many relationships and elements of data. A basic premise underlying the application of analytical procedures is that plausible relationships among data may reasonably be expected to exist and continue in the absence of known conditions to the contrary. Particular conditions that can cause variations in these relationships include, for example, specific unusual transactions or events, accounting changes, business changes, random fluctuations, or misstatements.

.03 Understanding financial relationships is essential in planning and evaluating the results of analytical procedures, and generally requires knowledge of the client and the industry or industries in which the client operates. An understanding of the purposes of analytical procedures and the limitations of those procedures is also important.

.04 Analytical procedures are used as a substantive test to obtain evidential matter about particular assertions related to account balances or classes of transactions. In some cases, analytical procedures can be more effective or efficient than tests of details for achieving particular substantive testing objectives.

.05 Analytical procedures involve comparisons of recorded amounts, or ratios developed from recorded amounts, to expectations developed by the auditor. The auditor develops such expectations by identifying and using plausible relationships that are reasonably expected to exist based on the auditor's understanding of the client and of the industry in which the client operates. Following are examples of sources of information for developing expectations:

- a. Financial information for comparable prior period(s) giving consideration to known changes
- b. Anticipated results—for example, budgets, or forecasts including extrapolations from interim or annual data
- c. Relationships among elements of financial information within the period
- d. Information regarding the industry in which the client operates—for example, gross margin information
- e. Relationships of financial information with relevant nonfinancial information

[.06-.08] [Paragraphs deleted.]

Analytical Procedures Used as Substantive Tests

.09 The auditor's reliance on substantive tests to achieve an audit objective related to a particular assertion¹ may be derived from tests of details, from analytical procedures, or from a combination of both. The decision about which procedure or procedures to use to achieve a particular audit objective is based on the auditor's judgment² on the expected effectiveness and efficiency of the available procedures. For significant risks of material misstatement, it is unlikely

that audit evidence obtained from substantive analytical procedures alone will be sufficient. (See paragraph .11 of AS 2301, *The Auditor's Responses to the Risks of Material Misstatement*.)

.10 The auditor considers the level of assurance, if any, he wants from substantive testing for a particular audit objective and decides, among other things, which procedure, or combination of procedures, can provide that level of assurance. For some assertions, analytical procedures are effective in providing the appropriate level of assurance. For other assertions, however, analytical procedures may not be as effective or efficient as tests of details in providing the desired level of assurance. When designing substantive analytical procedures, the auditor also should evaluate the risk of management override of controls. As part of this process, the auditor should evaluate whether such an override might have allowed adjustments outside of the normal period-end financial reporting process to have been made to the financial statements. Such adjustments might have resulted in artificial changes to the financial statement relationships being analyzed, causing the auditor to draw erroneous conclusions. For this reason, substantive analytical procedures alone are not well suited to detecting fraud.

.11 The expected effectiveness and efficiency of an analytical procedure in identifying potential misstatements depends on, among other things, (a) the nature of the assertion, (b) the plausibility and predictability of the relationship, (c) the availability and reliability of the data used to develop the expectation, and (d) the precision of the expectation.

Nature of Assertion

.12 Analytical procedures may be effective and efficient tests for assertions in which potential misstatements would not be apparent from an examination of the detailed evidence or in which detailed evidence is not readily available. For example, comparisons of aggregate salaries paid with the number of personnel may indicate unauthorized payments that may not be apparent from testing individual transactions. Differences from expected relationships may also indicate potential omissions when independent evidence that an individual transaction should have been recorded may not be readily available.

Plausibility and Predictability of the Relationship

.13 It is important for the auditor to understand the reasons that make relationships plausible because data sometimes appear to be related when they are not, which could lead the auditor to erroneous conclusions. In addition, the presence of an unexpected relationship can provide important evidence when appropriately scrutinized.

.14 As higher levels of assurance are desired from analytical procedures, more predictable relationships are required to develop the expectation. Relationships in a stable environment are usually more predictable than relationships in a dynamic or unstable environment. Relationships involving income statement accounts tend to be more predictable than relationships involving only balance sheet accounts since income statement accounts represent transactions over a period of time, whereas balance sheet accounts represent amounts as of a point in time. Relationships involving transactions subject to management discretion are sometimes less predictable. For example, management may elect to incur maintenance expense rather than replace plant and equipment, or they may delay advertising expenditures.

Availability and Reliability of Data

.15 Data may or may not be readily available to develop expectations for some assertions. For example, to test the completeness assertion, expected sales for some entities might be developed from production statistics or square feet of selling space. For other entities, data relevant to the assertion of completeness of sales may not be readily available, and it may be more effective or efficient to use the details of shipping records to test that assertion.

.16 Before using the results obtained from substantive analytical procedures, the auditor should either test the design and operating effectiveness of controls over financial information used in the substantive analytical procedures or perform other procedures to support the completeness and accuracy of the underlying information. The auditor obtains assurance from analytical procedures based upon the consistency of the recorded amounts with expectations developed from data derived from other sources. The reliability of the data used to develop the expectations should be appropriate for the desired level of assurance from the analytical procedure. The auditor should assess the reliability of the data by considering the source of the data and the conditions under which it was

gathered, as well as other knowledge the auditor may have about the data. The following factors influence the auditor's consideration of the reliability of data for purposes of achieving audit objectives:

- Whether the data was obtained from independent sources outside the entity or from sources within the entity
- Whether sources within the entity were independent of those who are responsible for the amount being audited
- Whether the data was developed under a reliable system with adequate controls
- Whether the data was subjected to audit testing in the current or prior year
- Whether the expectations were developed using data from a variety of sources

Precision of the Expectation

.17 The expectation should be precise enough to provide the desired level of assurance that differences that may be potential material misstatements, individually or when aggregated with other misstatements, would be identified for the auditor to investigate (see paragraph .20). As expectations become more precise, the range of expected differences becomes narrower and, accordingly, the likelihood increases that significant differences from the expectations are due to misstatements. The precision of the expectation depends on, among other things, the auditor's identification and consideration of factors that significantly affect the amount being audited and the level of detail of data used to develop the expectation.

.18 Many factors can influence financial relationships. For example, sales are affected by prices, volume and product mix. Each of these, in turn, may be affected by a number of factors, and offsetting factors can obscure misstatements. More effective identification of factors that significantly affect the relationship is generally needed as the desired level of assurance from analytical procedures increases.

.19 Expectations developed at a detailed level generally have a greater chance of detecting misstatement of a given amount than do broad comparisons. Monthly amounts will generally be more effective than annual amounts and comparisons by location or line of business usually will be more effective than

company-wide comparisons. The level of detail that is appropriate will be influenced by the nature of the client, its size and its complexity. Generally, the risk that material misstatement could be obscured by offsetting factors increases as a client's operations become more complex and more diversified. Disaggregation helps reduce this risk.

Investigation and Evaluation of Significant Differences

.20 In planning the analytical procedures as a substantive test, the auditor should consider the amount of difference from the expectation that can be accepted without further investigation. This consideration is influenced primarily by materiality and should be consistent with the level of assurance desired from the procedures. Determination of this amount involves considering the possibility that a combination of misstatements in the specific account balances, or class of transactions, or other balances or classes could aggregate to an unacceptable amount.

.21 The auditor should evaluate significant unexpected differences. Reconsidering the methods and factors used in developing the expectation and inquiry of management may assist the auditor in this regard. Management responses, however, should ordinarily be corroborated with other evidential matter. In those cases when an explanation for the difference cannot be obtained, the auditor should obtain sufficient evidence about the assertion by performing other audit procedures to satisfy himself as to whether the difference is a misstatement. In designing such other procedures, the auditor should consider that unexplained differences may indicate an increased risk of material misstatement. (See AS 2810.)

Documentation of Substantive Analytical Procedures

.22 When an analytical procedure is used as the principal substantive test of a significant financial statement assertion, the auditor should document all of the following:

- a. The expectation, where that expectation is not otherwise readily determinable from the documentation of the work performed, and factors considered in its development
- b. Results of the comparison of the expectation to the recorded amounts or ratios developed from recorded amounts

- c. Any additional auditing procedures performed in response to significant unexpected differences arising from the analytical procedure and the results of such additional procedures

[.23-.24] [Paragraphs deleted.]

Footnote (AS 2305 - Substantive Analytical Procedures):

¹ Assertions are representations by management that are embodied in financial statement components. See AS 1105, *Audit Evidence*.

² Reference to the judgment of the auditor has the same meaning as “professional judgment” as described in AS 1000, *General Responsibilities of the Auditor in Conducting an Audit*.

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2200 Auditing Internal Control Over Financial Reporting	+
2300 Audit Procedures in Response to Risks—Nature, Timing, and Extent	–
<u>AS 2301: The Auditor's Responses to the Risks of Material Misstatement</u>	
<u>AS 2305: Substantive Analytical Procedures</u>	
<u>AS 2310: The Auditor's Use of Confirmation</u>	
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2400 Audit Procedures for Specific Aspects of the Audit	+
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2800 Concluding Audit Procedures	+
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